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FACULTÉ DES SCIENCES ET TECHNOLOGIES

(FST)

Rapport du travail de Laboratoire #3

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1. Description des résultats de la tâche

1.1 Objectif du TD

Ce TD a pour but de nous apprendre à attribuer des adresses IP valides aux machines d'un réseau local, de comprendre la différence entre l'adressage IPv4 (32 bits, notation décimale pointée) et l'adressage IPv6 (128 bits, notation hexadécimale), puis de configurer ces adresses sur des hôtes (PC, laptop, serveur) dans Cisco Packet Tracer. Enfin, nous vérifions la connectivité entre les machines grâce à la commande ping.

Dans la partie « Travaux Dirigés », deux topologies en étoile sont à reproduire : l'une centrée sur un Hub (Hub2), l'autre centrée sur un Switch (2960-24TT, Switch3). Chacune est configurée en IPv4 puis en IPv6 selon le tableau d'adressage fourni.

1.2 Démarche suivie

1. Ouverture de Cisco Packet Tracer et création de la topologie à Hub : 1 Hub-PT, 1 LaptopPT (Admin) et 5 PC-PT (Client1 à Client5), reliés avec des câbles cuivre droits (Copper Straight-Through).

2. Configuration IPv4 statique de chaque machine : double-clic sur l'appareil, onglet Desktop, IP Configuration, option Static, puis saisie de l'adresse IPv4 et du masque 255.255.255.0.

3. Test de connectivité IPv4 depuis l'invite de commandes (Desktop > Command Prompt) avec la commande ping suivie de l'adresse de la machine cible.

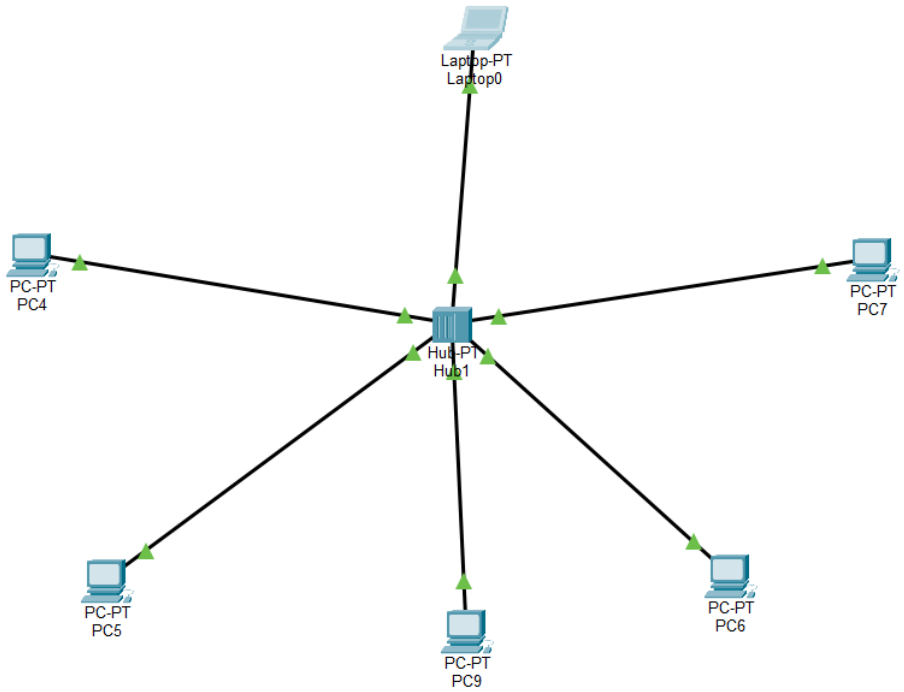
4. Création de la topologie à Switch : 1 Switch 2960-24TT, 1 Laptop-PT (Admin), 6 PC-PT (Client1 à Client6) et 1 Server-PT (Server Mail), puis configuration IPv4 et tests de ping.

5. Ajout d'une adresse IPv6 statique avec préfixe /64 sur chaque machine (même fenêtre IP Configuration, section IPv6 Configuration), puis tests avec ping suivi de l'adresse IPv6.

6. Prise des captures d'écran de chaque configuration et de chaque test, puis rédaction du rapport et hébergement sur GitHub avec Git

2. Résultats de l'exécution des commandes

Exercice 1 – Topologie à Hub, adressage IPv4




```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.14

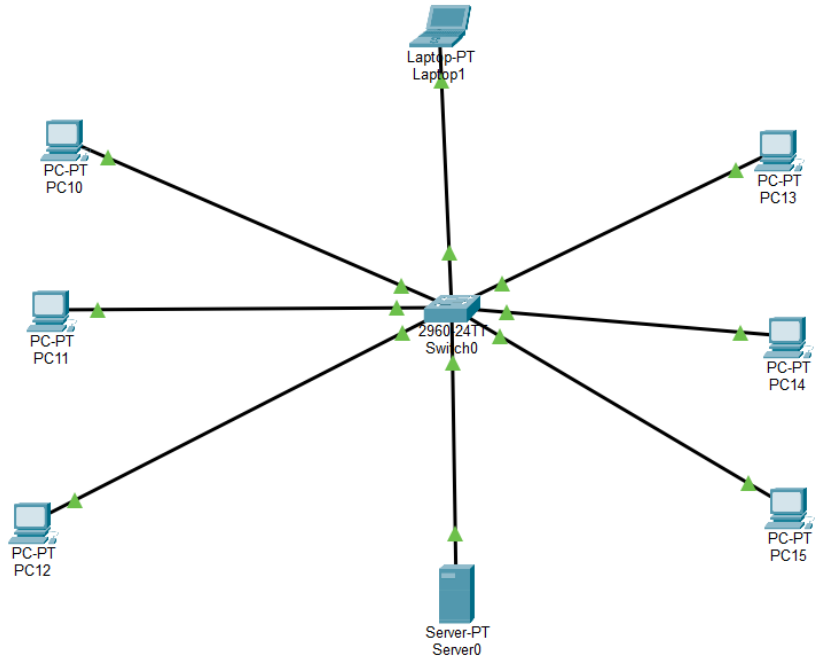
Pinging 192.168.1.14 with 32 bytes of data:

Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Exercice 2 – Topologie à Switch, adressage IPv4

Topologie reproduite : Switch3 (2960-24TT) au centre, relié à Admin, Client1 à Client6 et Server Mail




```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

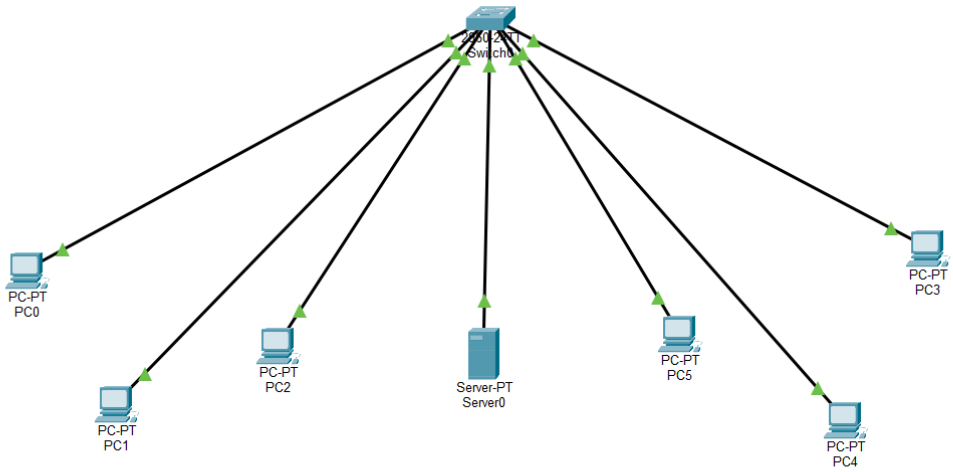
C:\>ping 192.168.1.14

Pinging 192.168.1.14 with 32 bytes of data:

Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Exercice 3 – Topologie etoile, adressage IPv4




```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=2ms TTL=128
Reply from 192.168.1.12: bytes=32 time=9ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 5ms

C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

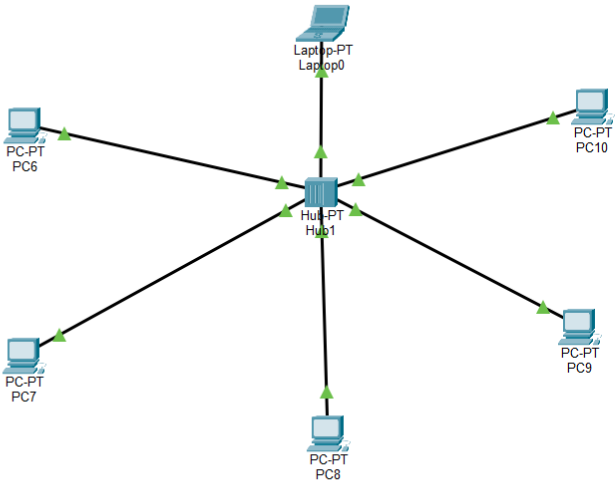
C:\>ping 192.168.1.14

Pinging 192.168.1.14 with 32 bytes of data:

Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time=1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Exercice 4 – Topologie à Hub, adressage IPv6



Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::1

Pinging 3000:159:1::1 with 32 bytes of data:

Reply from 3000:159:1::1: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=6ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=6ms TTL=128

Ping statistics for 3000:159:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 6ms, Average = 5ms

C:\>ping 3000:159:1::2

Pinging 3000:159:1::2 with 32 bytes of data:

Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=9ms TTL=128

Ping statistics for 3000:159:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 3000:159:1::3

Pinging 3000:159:1::3 with 32 bytes of data:

Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=7ms TTL=128

Ping statistics for 3000:159:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 1ms

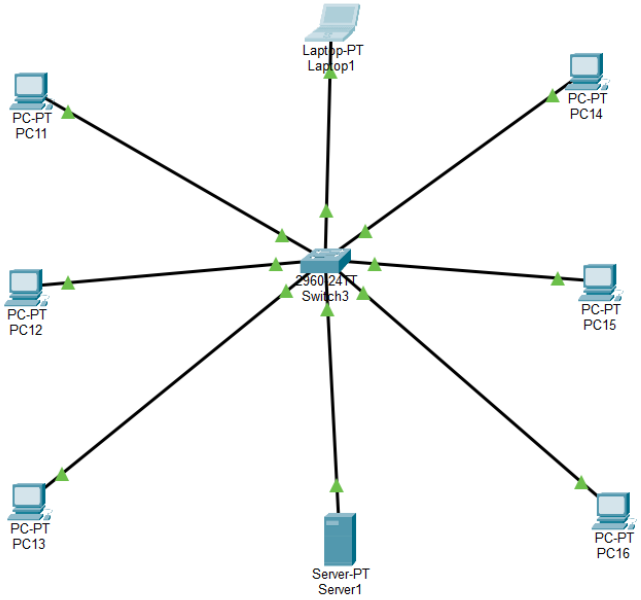
C:\>ping 3000:159:1::4

Pinging 3000:159:1::4 with 32 bytes of data:

Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Exercice 5 – Topologie à Switch, adressage IPv6



Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::1

Pinging 3000:159:1::1 with 32 bytes of data:

Reply from 3000:159:1::1: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=6ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=6ms TTL=128

Ping statistics for 3000:159:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 6ms, Average = 5ms

C:\>ping 3000:159:1::2

Pinging 3000:159:1::2 with 32 bytes of data:

Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=9ms TTL=128

Ping statistics for 3000:159:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 3000:159:1::3

Pinging 3000:159:1::3 with 32 bytes of data:

Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=7ms TTL=128

Ping statistics for 3000:159:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 1ms

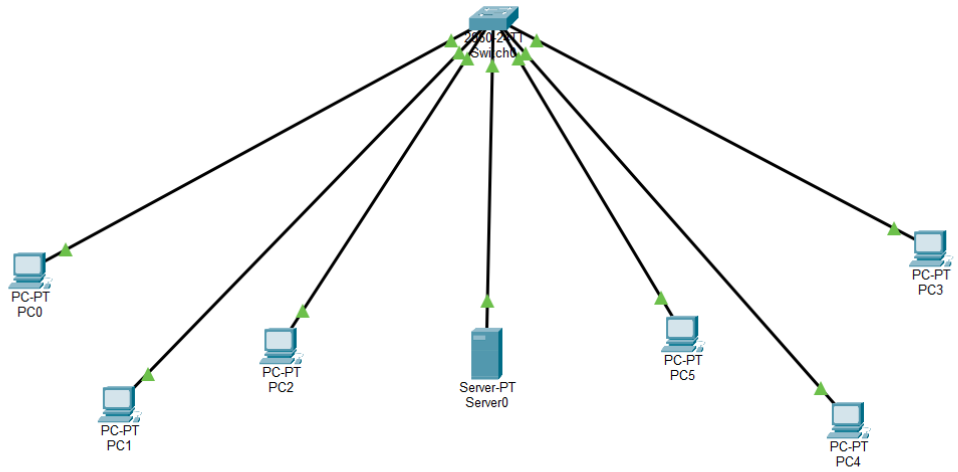
C:\>ping 3000:159:1::4

Pinging 3000:159:1::4 with 32 bytes of data:

Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Exercice 6 – Topologie hybride, adressage IPv6




```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=2ms TTL=128
Reply from 192.168.1.12: bytes=32 time=9ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 5ms

C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.14

Pinging 192.168.1.14 with 32 bytes of data:

Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time=1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

3. Conclusions

3.1 Ce que j’ai appris

• Attribuer une adresse IPv4 valide a une machine.

• Comprendre l'adressage IPv6

• Configurer des adresses IP statiques sur des PC, un laptop et un serveur dans Cisco Packet Tracer.

• Verifier la connectivite avec ping (IPv4 et IPv6) et lire ses resultats : paquets envoyes/reçus, pourcentage de perte, temps de reponse.

3.2 Difficultes rencontrees et solutions

• Prefixe IPv6 invalide : « 3000:is9:1::/64 » contient des caracteres non hexadecimaux. Solution : utilisation du prefixe valide 3000:159:1::/64